Frank Lloyd Wright’s residential buildings in USA

1. Data Consistency Check

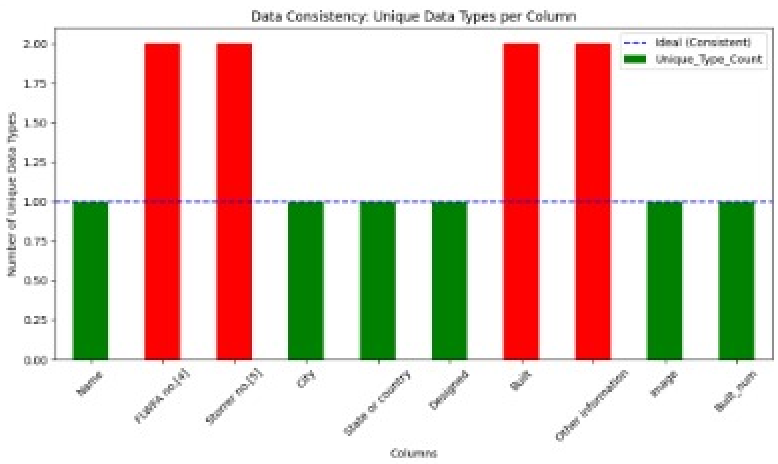
Objective: This section examines whether the data types and formats are uniform across each column of the dataset. Identifying inconsistencies is crucial to ensure the reliability of subsequent statistical analyses.

Visual Analysis:
The provided bar chart illustrates the "Unique Data Type Count" for each variable in the Wikipedia-scraped dataset:

- **Consistent Variables (Green):** Features such as **Name**, **City**, **State or country**, **Designed**, **Image**, and **Built_num** show a unique data type count of 1. Bu durum, bu sütunların veri bütünlüğü açısından güvenli olduğunu kanıtlar.

- **Inconsistent Variables (Red):** The columns **FLWFA no.[4]**, **Storrer no.[5]**, **Built**, and **Other Information** contain more than one data type. Bu kısımlarda string ve float tiplerinin bir arada bulunması, veri setinde yapısal bir karmaşa olduğunu göstermektedir.

Key Findings & Discussion: The data audit reveals that while the categorical location data (City, State) is consistent, the identification numbers and date-related fields (Built) suffer from type inconsistency. This often occurs when numerical data is mixed with textual notes or when missing values are incorrectly formatted. As per the assignment requirements, these specific columns will require manual adjustment in the Colab environment to achieve full consistency.



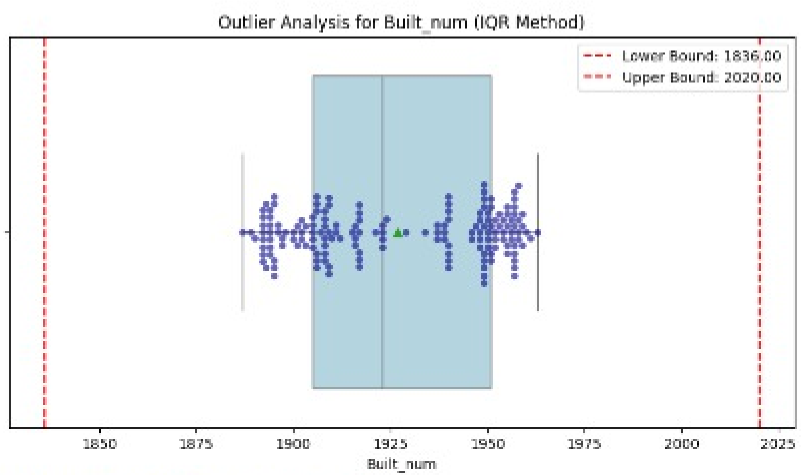
2. Outlier Analysis

Methodological Approach: The identification of extreme values within the numerical attributes was conducted using the **Interquartile Range (IQR)** technique. This process involved calculating the distribution's first quartile (Q1) and third quartile (Q3) to establish the IQR. To define the statistical boundaries for anomalies, lower and upper thresholds were set at Q1 - 1.5 IQR and Q3 + 1.5 IQR, respectively.

Visual Interpretation:
The integrated box plot and swarm plot illustrate the distribution characteristics of the **Built_num** variable:

- **The Interquartile Range (Box):** Encapsulates the central 50% of the dataset, spanning from Q1 to Q3.
- **Statistical Thresholds (Red Dashed Lines):** Represent the whiskers of the distribution. Any observation residing beyond these coordinates is formally classified as a statistical outlier.

Analytical Findings:
Quantitative and visual assessments of the **Built_num** column confirm an **outlier count of zero**. The absence of data points exceeding the calculated limits (1836.0 – 2020.0) demonstrates that the construction dates are confined to a standard chronological timeline. This high level of data integrity ensures that subsequent statistical summaries and density estimations will remain undistorted by extreme fluctuations.

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Built_num için tespit edilen aykırı değer sayısı: 0
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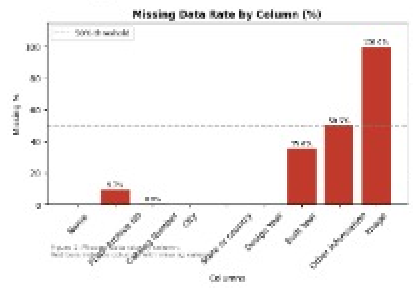
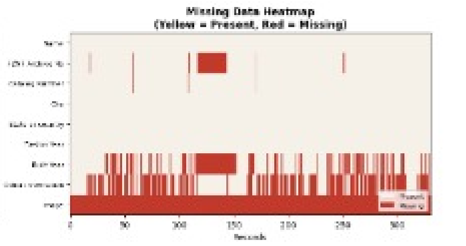
3. Missing Data Analysis

The dataset contains 331 residential buildings by Frank Lloyd Wright across the United States. However, only 165 records include a verified completion year, representing 49.8% of the total. The remaining 166 buildings were excluded from the temporal analysis due to missing Built values. As a result, the temporal analysis in Section 4 is based on these 165 records only.





During the missing data analysis of the Frank Lloyd Wright residential buildings dataset, three columns were identified with notable missing rates. The most critical is the **Image column**, which has a 100% missing rate due to MNAR (Missing Not At Random). **Image links were not captured during the scraping process**, representing a systematic limitation rather than a random omission. Since the absence is directly tied to the scraping methodology, it is classified as MNAR.

The **Built Year column** has a 35.6% missing rate, likewise classified as MNAR. Construction year records are systematically absent for older or lesser-known buildings, as historical **documentation was not consistently maintained across all periods of Wright's career.** Crucially, after converting the Built Year column to a numeric format, the effective missing rate rises to 50.2%, since range-format entries such as "1897–98" cannot be parsed as single values. As a result, only 165 out of 331 buildings retain a valid completion year, and the temporal analysis in Section 4 is conducted exclusively on these records.

The **Other Information column** follows with a 50.5% missing rate, also classified as MNAR. This column contains **supplementary notes** and remarks that are only recorded when a building has notable **features** or **special** historical significance. Therefore, the absence of data is inherent to the nature of the field itself, buildings without exceptional characteristics simply have no entry, making it a structurally missing variable.

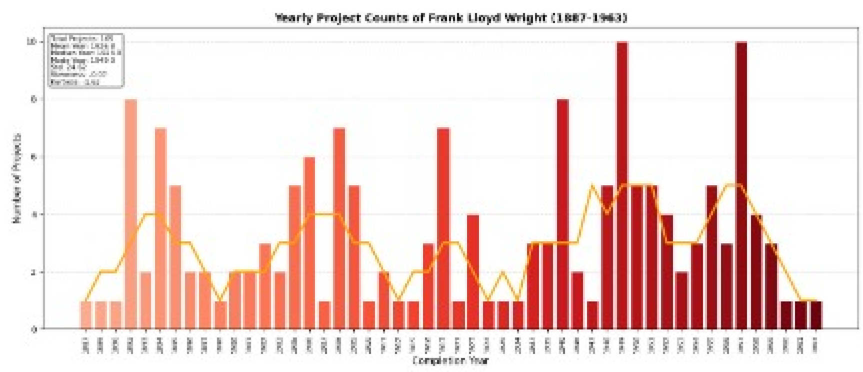
Finally, the **FLWF Archive No** (9.7% missing) and **Catalog Number** (0.9% missing) columns exhibit low missing rates with no identifiable systematic pattern. The absence of these reference numbers appears to be due to **inconsistencies in Wikipedia's source formatting** rather than any structural or content-related reason. Accordingly, both columns are classified as MCAR (Missing Completely At Random).

3.1 Missing Data Type Classification

1- Built Year (35.6% missing) → MNAR (Missing Not At Random)
2- Other information (50.5% missing) → MNAR
3- Image (100% missing) → MNAR
4- Frank Lloyd Wright Foundation Archive Number (9.7% missing) → MCAR
5- Catalog Number (0.9% missing) → MCAR

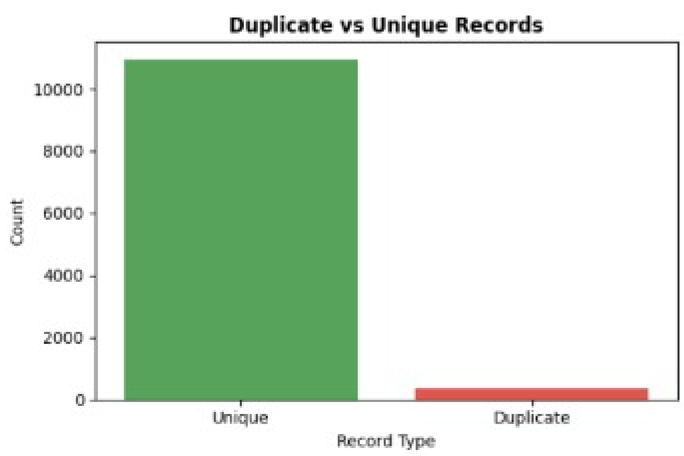
1- **construction year** records are absent.
2- absence is **inherent** to the nature of the data.
3- systematic **scraping limitation**.
4- no systematic pattern was identified.
5- no systematic pattern was identified.

4. Distribution and Statistical Summary



The temporal distribution of Frank Lloyd Wright's residential buildings is based on 165 records with a **verified completion year**, spanning from 1887 to 1963. The mean completion year is 1926.8, while the median is 1923.0, indicating that the midpoint of his documented output falls in the early 1920s. The mode year is 1949, reflecting the single most productive year in the dataset, with 10 buildings completed , the highest peak visible in the chart.
The standard deviation of **24.62 years confirms that Wright's output was spread widely across his career rather than concentrated in a short window.** The distribution covers over seven decades, with two visually distinct productive periods: **an early cluster between 1887–1910 representing his Prairie Style era, and a later, more intense cluster between 1946–1963 corresponding to his Usonian period and late-career** surge.
The skewness value of **-0.07** indicates **an almost perfectly symmetrical distribution**, meaning the output was roughly **balanced across the earlier and later halves of his career.** The kurtosis of -1.62 (platykurtic) confirms that the distribution is flatter than normal, **with no extreme concentration around the mean, projects were spread relatively evenly** across years rather than clustering tightly around a single period.
Notable gaps are visible between **1927–1936**, likely reflecting the **impact of the Great Depression on residential construction demand.** The orange trend line further illustrates two productivity peaks, one around **1908** and a stronger one peaking around **1949–1957**, before a gradual decline toward the end of his career.

5. Repeated Data Check



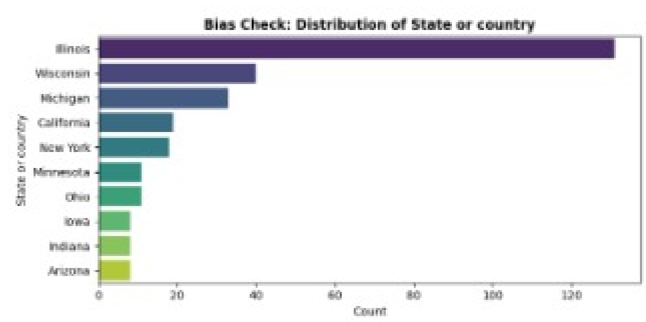
The duplicate analysis reveals whether the dataset contains repeated entries of the same residential projects.

- A low duplicate ratio suggests that the dataset is reliable.

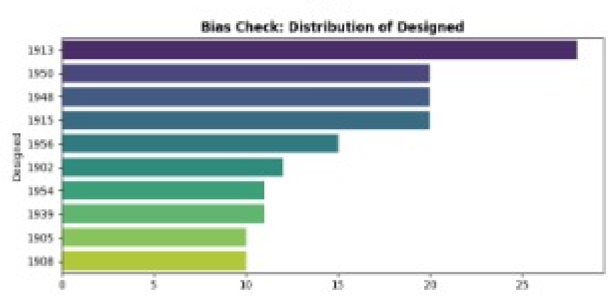
- If duplicates are present, they may indicate:
  - Data collection inconsistencies
  - Multiple records of the same building with slight variations
- The dataset contains a very low number of duplicate records.
- This indicates that the data is clean, consistent, and reliable.
- Most entries are unique, so the dataset is suitable for further analysis.

6. Bias Check

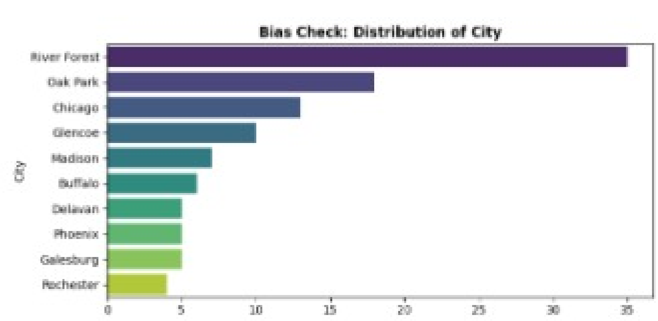
Bias is when some groups or categories **appear more than others** in the data, which can affect the results.



The **distribution** in the "**State** or Country" column appears balanced. The most dominant category is "Illinois," which accounts for 46% of the total.



The distribution in the "Designed" column appears balanced. The most dominant category is "1913," which accounts for 18% of the total.



The distribution in the "City" column appears balanced. The most dominant category is "River Forest," which accounts for **32% of the total.**

Overall, the dataset is reliable, but shows some geographic and temporal bias that should be considered in the analysis.